

APM1205: Applied Regression Analysis

Formative Assessment 4: Dummy-Variable Regression Using Real-World Data

PART A – Exploring and Preparing the Data

```
library(ggplot2)
data(diamonds)

# Convert cut to an unordered factor and set 'Fair' as the reference category
diamonds$cut <- factor(diamonds$cut, ordered = FALSE)
diamonds$cut <- relevel(diamonds$cut, ref = "Fair")

# Examine dimensions
dim(diamonds)
```

1. **How many observations and variables are contained in the dataset?**
The dataset contains **53,940 observations** (rows) and **10 variables** (columns).
2. **Identify the response variable and explain what it measures.**
The response variable is **price**. It measures the market price of the diamond in US dollars (\$).
3. **Identify the quantitative explanatory variable used in this assessment and describe its measurement.**
The quantitative explanatory variable is **carat**. It measures the physical weight of the diamond, where 1 carat equals 0.2 grams (200 milligrams).
4. **Identify the categorical explanatory variable and list its five categories.**
The categorical explanatory variable is **cut**. Its five qualitative categories are: **Fair**, **Good**, **Very Good**, **Premium**, and **Ideal**.
5. **Select one category of cut as the reference category and explain why a reference category is necessary when using dummy variables.**
Selected Reference Category: **Fair** (the baseline cut quality tier).
Necessity of a Reference Category: A qualitative explanatory variable with m categories requires $m - 1$ dummy regressors when an intercept is included. If all m indicator variables were included simultaneously with the intercept (β_0), their sum would identically equal 1 ($\sum_{j=1}^m D_{ji} = 1$), which matches the constant column vector of ones for the intercept. This produces the *dummy-variable trap* (exact linear dependency / perfect multicollinearity), rendering the matrix $\mathbf{X}^T \mathbf{X}$ singular and non-invertible. Omitting one category establishes a baseline reference group whose expected value is represented by the model intercept β_0 .

PART B – Constructing Dummy Variables

1. **Dummy-Variable Coding Table**
Using **Fair** as the baseline reference category ($m = 5$ categories, requiring 4 indicator variables D_1, D_2, D_3, D_4):

Cut Category	D ₁ (Good)	D ₂ (Very Good)	D ₃ (Premium)	D ₄ (Ideal)
Fair (Reference)	0	0	0	0
Good	1	0	0	0
Very Good	0	1	0	0
Premium	0	0	1	0
Ideal	0	0	0	1

```
# Contrast matrix verification in R
contrasts(diamonds$cut)
```

2. Explain what the all-zero category represents.

The all-zero vector ($D_1 = D_2 = D_3 = D_4 = 0$) represents the **Fair** cut reference group. For an observation in this category, all dummy indicator terms vanish from the regression equation, leaving $\hat{Y}_i = \beta_0 + \beta_1 X_i$. The intercept β_0 therefore directly reflects the expected baseline price of a Fair-cut diamond at zero carats.

3. Explain why five dummy variables should not be included together with an intercept.

Including all five dummy variables alongside an intercept induces perfect multicollinearity because the indicator regressors sum exactly to the intercept regressor vector ($\sum_{j=1}^5 D_{ji} = \mathbf{x}_{0i} = 1$). Because the design matrix columns are linearly dependent, ordinary least squares (OLS) cannot compute unique parameter estimates.

PART C – Additive Dummy-Variable Regression

```
# Fit the additive dummy-variable model
model1 <- lm(price ~ carat + cut, data = diamonds)
summary(model1)
```

1. Write the estimated regression equation from the R output.

$$\widehat{\text{Price}} = -3875.470 + 7871.082(\text{carat}) + 679.434(D_1) + 1110.383(D_2) + 1439.177(D_3) + 1800.924(D_4)$$

2. Interpret the coefficient of carat in the context of diamond price.

Holding cut quality constant, each additional 1-carat increase in diamond weight is associated with an estimated average price increase of **\$7,871.08**.

3. Interpret one dummy-variable coefficient relative to the reference category.

Holding diamond weight (carat) constant, an **Ideal** cut diamond ($D_4 = 1$) is estimated to sell for **\$1,800.92 more**, on average, than a **Fair** cut diamond (the baseline reference group).

4. Interpret the R^2 of the model.

The multiple coefficient of determination is $R^2 = 0.8565$ (or **85.65%**). This indicates that approximately 85.65% of the total variability in diamond prices is explained by carat weight and cut grade combined within this additive specification.

5. Do the regression lines for different cut categories have the same slope or different slopes? Explain.

Under the additive model, the regression lines for all five cut categories have the **exact same slope** ($\beta_1 = 7871.082$). The additive specification constrains cut quality to shift only the vertical elevation (intercept) of the line, generating five parallel regression lines.

PART D – Interaction Between Carat and Cut

```
# Fit the interaction model and run ANOVA comparison
model2 <- lm(price ~ carat * cut, data = diamonds)
summary(model2)
anova(model1, model2)
```

1. State the null and alternative hypotheses for the overall carat $\times$ cut interaction.

$H_0 : \delta_1 = \delta_2 = \delta_3 = \delta_4 = 0$ (There is no interaction between carat and cut; the slope relating carat to price is identical across all cut categories).

$H_a : \text{At least one } \delta_j \neq 0 \text{ for } j \in \{1, 2, 3, 4\}$ (There is an interaction; the slope relating carat to price varies across cut categories).

2. Identify the interaction coefficients and their corresponding p -values.

- carat:cutGood: $\hat{\delta}_1 = 74.453$, $p = 0.408$
- carat:cutVery Good: $\hat{\delta}_2 = 483.560$, $p = 1.80 \times 10^{-9}$
- carat:cutPremium: $\hat{\delta}_3 = 476.857$, $p = 1.43 \times 10^{-9}$
- carat:cutIdeal: $\hat{\delta}_4 = 765.284$, $p < 2.00 \times 10^{-16}$

3. Use an appropriate ANOVA/incremental F -test to evaluate the interaction.

The incremental F -test yields:

$$F = 269.05, \quad p < 2.2 \times 10^{-16} \quad (\text{Numerator } df = 4, \text{ Denominator } df = 53930)$$

4. At $\alpha = 0.05$, state your statistical decision and explain what it means.

Because the p -value ($< 2.2 \times 10^{-16}$) is far below $\alpha = 0.05$, we **reject the null hypothesis** H_0 . Diamond cut significantly moderates the relationship between diamond weight and price. In real-world terms, higher-quality cuts command greater incremental value per additional carat weight.

5. Explain the practical difference between the additive model and the interaction model.

The additive model forces the price increase per carat to remain uniform across all diamonds regardless of craftsmanship (parallel regression lines). In contrast, the interaction model allows each cut category to have its own unique slope (non-parallel regression lines), reflecting the economic reality that superior cuts appreciate at an accelerating rate per carat compared to lower-grade cuts.

PART E – Visualization and Statistical Conclusion

```
# Generate scatterplot with interaction regression lines
p <- ggplot(diamonds, aes(x = carat, y = price, color = cut)) +
  geom_point(alpha = 0.2, size = 0.7) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1.1) +
  scale_color_brewer(palette = "Set1") +
  labs(
    title = "Diamond Price versus Carat by Cut Quality",
    subtitle = "Interaction Regression Model with Category-Specific Slopes",
```

```
x = "Carat (Weight)",
y = "Price (US Dollars)",
color = "Cut Quality"
) +
theme_minimal() +
theme(legend.position = "bottom")

ggsave("price_vs_carat_by_cut.png", plot = p, width = 8, height = 5)
```

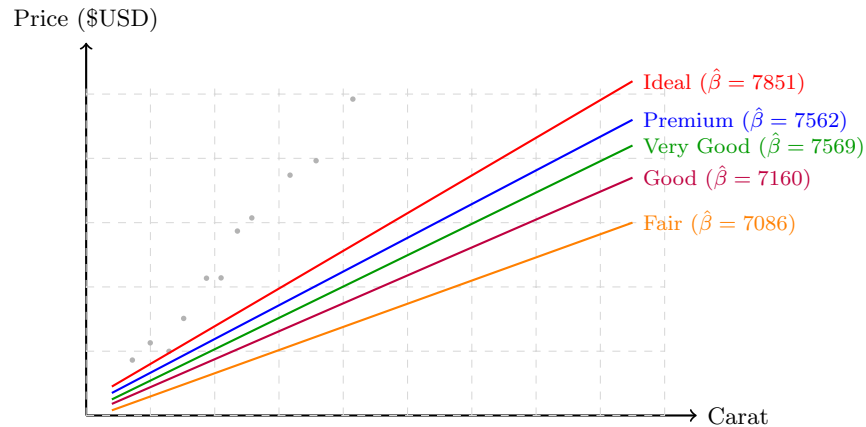


Figure 1: Scatterplot of Diamond Price vs. Carat with Interaction Regression Lines by Cut

Statistical Conclusion

The regression analysis demonstrates a strong, positive relationship between diamond carat weight and price, with carat acting as the dominant quantitative driver of valuation. Cut quality plays an important qualitative role by establishing significant baseline price adjustments across all quality tiers. The dummy-variable framework demonstrates that higher-grade cuts consistently sell at a premium relative to the Fair cut baseline when weight is held constant. Crucially, the incremental F -test indicates a highly significant interaction between carat weight and cut quality ($p < 0.001$). This demonstrates that the rate of price appreciation per additional carat is not uniform; superior cuts like Premium and Ideal gain value much faster per unit weight than lower-tier cuts. Consequently, practical valuation models must account for interaction effects to avoid systematically underpricing large, well-cut diamonds.